From Collisions to Recoveries: A Comparative Study of MARSALA Random Access

CESE4055 Group Project 2
Ad Hoc and Sensor Networks

***Abstract*—Satellite random access must recover packets when several users transmit in the same time slot. This paper compares slotted ALOHA, contention resolution diversity slotted ALOHA (CRDSA), and CRDSA with multi-replica decoding using correlation-based localisation (MARSALA). We explain how repetition creates both diversity and additional contention, then examine the effects of replica count, signal-to-noise ratio (SNR), decoding threshold, and load. At an equal arrival load of 75 packets per 100-slot frame, the model delivers 0.356, 0.502, and 0.750 packets per slot for slotted ALOHA, two-replica CRDSA, and two-replica MARSALA, respectively. The receiver is an oracle-assisted concept model, so these values illustrate mechanisms rather than predict physical-link performance.**

***Index Terms*—random access, slotted ALOHA, CRDSA, MARSALA, successive interference cancellation, satellite communications**

I. Introduction

In a satellite return link, many terminals share a finite set of time slots. Without scheduling, each terminal chooses its own transmission slot and collisions are unavoidable. Slotted ALOHA treats a collision as a loss. CRDSA instead sends several copies of each packet and uses successive interference cancellation (SIC) to peel decoded packets from other slots [1]. MARSALA retains the CRDSA transmitter but adds a receiver step: it localises and combines collided replicas when SIC has no singleton left to decode [2].

The important question is not only which receiver delivers more packets, but *when* its extra complexity pays off. We compare the three receivers at equal packet demand, study four adjustable parameters, and distinguish the idealised model's predictions from a practical satellite link.

II. Access and Decoding Mechanisms

Let N_s be the number of slots, M the number of users, and R the replicas per packet. Packet arrival load and replica load are different:

$$U = \frac{M}{N_s}, \qquad G = \frac{MR}{N_s} = UR. \tag{1}$$

Equal G across different R does not imply equal user demand. We hold U fixed for the parameter comparison.

A. Slotted ALOHA

Each user places one packet in one randomly chosen slot. A packet is received only if no other user chooses that slot. For independent choices, the finite frame throughput is

$$T_{\mathrm{SA}} = \frac{M}{N_s}\left(1 - \frac{1}{N_s}\right)^{M-1} \approx Ue^{-U}. \tag{2}$$

The large-frame peak is $1/e \approx 0.368$ packets per slot. At $M = 75$ and $N_s = 100$, (2) predicts approximately 0.355, close to the measured 0.356.

B. CRDSA and SIC

CRDSA transmits R copies in different slots. The receiver decodes a singleton, cancels that packet's other copies, and repeats. A collision can become a singleton after cancellation. If the remaining packets form a loop with no singleton, this peeling process reaches a deadlock. Repetition helps recovery but also raises G and therefore creates more collisions.

C. MARSALA at Deadlock

MARSALA cross-correlates collided slot signals to locate replicas of one packet, aligns their phases, combines them, and retries decoding. A newly decoded packet can restart SIC. Under equal-power replicas and independent interference, the approximate combined signal-to-interference-plus-noise ratio (SINR) is

$$\mathrm{SINR}_{\mathrm{comb}} \simeq \frac{R\rho}{1 + (k-1)\rho}, \tag{3}$$

where ρ is per-replica SNR and k is collision depth. The relation explains why more copies or higher SNR can help, whereas deeper collisions or a stricter decoding threshold can prevent recovery. It is an approximation, not a complete link budget.

III. Model and Comparison Method

Each simulated frame contains $N_s = 100$ slots and 256 complex samples per slot. Replicas have independent random phases, small amplitude variation, and additive Gaussian noise. A packet decodes when its modelled SINR exceeds a threshold γ. The default is $R = 2$, per-replica SNR -2.2 dB, and $\gamma = 0.61$. The controlled parameter study uses $M = 75$ for every receiver and averages 60 frames per point. Throughput is delivered packets divided by N_s; frame loss is the undelivered fraction of the offered packets.

The fixed example frame illustrates the sequence: slotted ALOHA recovers 31/75 packets, CRDSA recovers 62/75, and MARSALA recovers 75/75. The 13 additional packets follow replica combination and renewed SIC. This example is useful for explaining the algorithm; averaged results below describe the parameter effects.

TABLE I
THROUGHPUT AT EQUAL PACKET LOAD $U = 0.75$

R	Slotted ALOHA	CRDSA	MARSALA
1	0.356	0.356	0.356
2	0.356	0.502	0.750
3	0.356	0.631	0.750
4	0.356	0.428	0.750

IV. PARAMETER EFFECTS

A. Number of Replicas

Table I shows diminishing returns from repetition. At $R = 2$, SIC raises throughput from the slotted ALOHA baseline of 0.356 to 0.502, and combining raises it to the offered-demand ceiling of 0.750. At $R = 3$, CRDSA improves to 0.631. At $R = 4$, its throughput falls to 0.428: the extra transmissions make slots more crowded. MARSALA stays at the ceiling in this idealised study, so further copies use more channel resources without delivering more packets. For $R = 2$, the corresponding frame loss is about 33% under CRDSA and approximately zero under MARSALA.

B. Per-Replica SNR

With $R = 2$, raising SNR from -6 to -4 to -2.2 dB increases MARSALA throughput from 0.502 to 0.633 to 0.750 packets per slot. The ALOHA and CRDSA curves barely move because the singleton decoding stand-in already succeeds. SNR mainly determines whether a *combined collision* crosses the decoding threshold. Above -2.2 dB, throughput cannot rise further because all offered packets are recovered in this experiment.

C. Required Decoding SINR

At the default SNR, increasing γ from 0.61 to 0.8 to 1.0 reduces MARSALA throughput from 0.750 to 0.640 to 0.502. At $\gamma = 1.0$, combining adds no recoveries over CRDSA. A stricter threshold is a stand-in for a less capable code or a stronger reliability requirement. It excludes borderline combined packets; the model does not quantify any coding gain or error-rate benefit from that stricter requirement.

D. Offered Load

As U grows, singleton slots become rarer and ALOHA loss rises. SIC keeps CRDSA useful beyond ALOHA's peak, while MARSALA can resolve some deadlocks. At high occupancy, collision depth increases and the receiver must search more candidate slots. Fig. 2 plots delivered packets against replica load G; curves for different R should not be compared as equal-demand experiments at the same horizontal coordinate.

V. DELAY, LOSS, AND COMPLEXITY

Undelivered packets in the delay experiment retry in new slots the next frame and are dropped after 20 attempts. Mean delay is calculated only over delivered packets, so it must be read alongside loss. At $G = 2.5$ with $R = 2$, CRDSA's mean delay is 4.62 frames and MARSALA's is 1.04

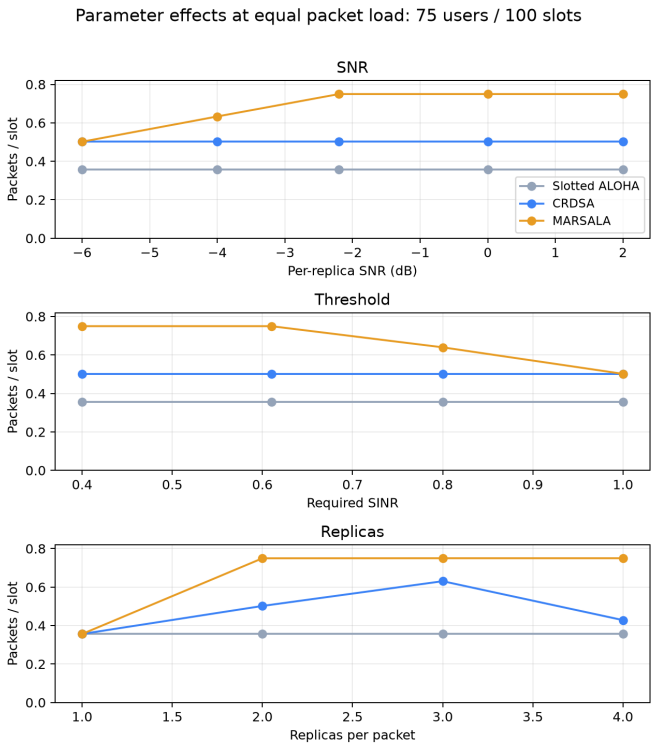


Fig. 1. Effect of SNR, decoding threshold, and replica count at equal arrival load $U = 0.75$. Each point averages 60 frames.

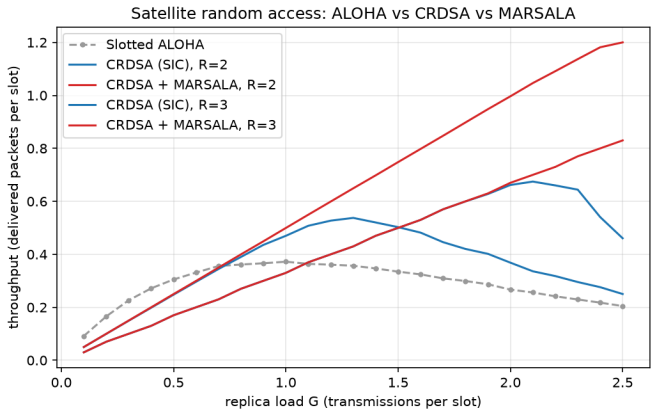


Fig. 2. Throughput versus replica load for the three receiver families. The same G corresponds to different user counts when R differs.

frames in this model. Unlocking a SIC cascade can therefore reduce retransmission time. At equal demand $U = 0.75$, the measured frame-loss fractions are approximately 52.5% for slotted ALOHA, 33.1% for two-replica CRDSA, and 0% for two-replica MARSALA. Loss supplies essential context for the mean delay, which excludes packets that were dropped.

MARSALA incurs extra receiver work: each deadlock requires slot correlations, replica localisation, phase alignment, and combining. At $G = 2.5$ with $R = 2$, this implementation averages about 11,353 correlations per frame.

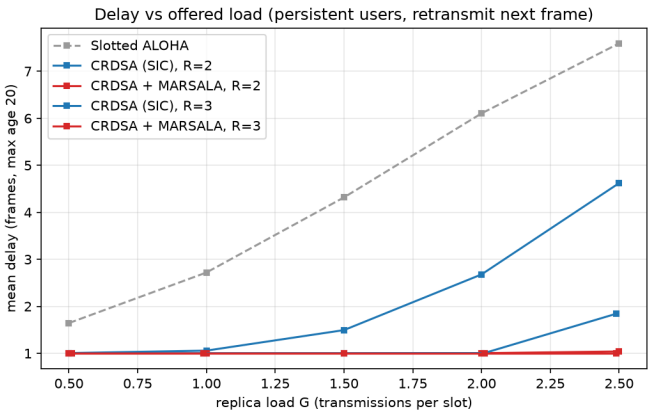


Fig. 3. Mean delivered-packet delay under immediate next-frame retransmission. Dropped packets are excluded from the mean.

VI. Scope and Limitations

This is an *oracle-assisted concept simulation*. The decoder reads true slot occupancy, packet symbols, and amplitudes from the frame generator. It does not perform blind packet identification, imperfect channel estimation, a complete error-correcting code, or physical signal subtraction. The modelled threshold is a surrogate for decoding success, and the delay model omits feedback and propagation overhead. Thus the figures illustrate how the mechanisms interact, not validated satellite-link performance. In particular, the very high two-replica MARSALA throughput under heavy load should not be quoted as a real-world prediction.

VII. Conclusion

Slotted ALOHA loses collisions; CRDSA recovers some of them through replication and SIC; MARSALA can combine collided replicas to restart SIC. The controlled study shows that SNR and decoding threshold govern whether combining helps, while replica count balances diversity against added contention. The gain costs receiver correlations and depends on channel knowledge that this concept model idealises. These tradeoffs, rather than a single peak-throughput number, are the main lesson of the comparison.

References

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[2] H.-C. Bui, K. Zidane, J. Lacan, and M.-L. Boucheret, "A Multi-Replica Decoding Technique for Contention Resolution Diversity Slotted Aloha," in *Proc. IEEE VTC-Fall*, 2015, doi: 10.1109/VTCFall.2015.7390798.